SQL BASIC COMMANDS

DDL COMMAND

CREATE
ALTER
DROP
TRUNCATE
RENAME

DML COMMANDS

INSERT
UPDATE
DELETE
MERGE

DQL COMMANDS

SELECT

DDL COMMANDS

CREATE

BASICS
--DATABASE
--TABLES
--INDEX
--VIEW

ADVANCE
--SCHEMA
--SEQUENCE
--FUNCTION
--PROCEDURE
--TRIGGER

SECURITY&ACCESS
--USER
--ROLE

Other Useful Objects
--MATERIALIZED VIEW
--SYNONYM
--TYPE

ALTER

--ADD COLUMNS
--MODIFY COLUMNS
--DROP COLUMNS
--RENAME COLUMN
--ADD CONSTRAINTS
--COLUMN POSITION CHANGE

DROP

--DROP IS USED TO DELETE WHOLE OBJECTS
-- IN SIMPLE LAUNGUAGE JITNI BHI CHIZE HUM CREATE KRTE HAI WO SAB COMPLITLY DELETE KRNA

TRUNCATE

--TRUNCATE SE TABLE KA PURA DATA DELETE HOTA HAI COLUMNS NAME OR STRUCTER TABLE KA WESA HI REHTA HAI
--SIRF SARI ROWS DELETE HOGI
--TRUCATE SE SARA DATA UDEGA ESA NHI KI EK LINE YA EK COLUMNS HI DELETE HOGI.

RENAME

--RENAME KR SKTE HAI
COLUMNS
INDEX
TABLES
DATABASE

DML COMMANDS

INSERT

--INSERT DATA INTO COLUMNS

METHOD
--INSERT INTO COLUMN_NAME()
VALUES()

UPDATE

--UPDATE THE EXICTING DATA FROM COLUMN

--ITS ONLY CHANGE THE VALUE NOT CREAT VALUE OR DATA

METHOD
--UPDATE TABLE_NAME
SET COLUMN_NAME="VALUE"
WHERE COLUMN_NAME = VALUE

DELETE

--DELETE VALUES FROM COLUMN AND WE HAVE TO GIVE LOCATION FOR DELETE ROW VALUES.

MERGE

--MERGE DO TABLES ME USE HOTA HAI LIKE AGAR KISI EK TABLE ME VALUES HAI TO WO UPDATE KREGA OR NAHI HAI TO NEW VALUE INSERT KREGA

--NICHE DETAILS ME HAI

DQL COMMANDS

SELECT <----- SELECT IS ONLY DQL COMMAND WHICH IS USE FOR READ AND FATCH THE DATA

METHODS FOR WRITING SELECT

1. SELECT * FROM TABLE_NAME; -----> FOR READ ALL THE DATA FROM TABLE
2.SELECT COLUMN_NAME FROM TABLE_NAME; ----->FOR READ DATA FROM SPECIFIC COLUMN FROM TABLE
3.SELECT COLUMN1 , COLUMN2, ETC FROM TABLE_NAME ----->FOR READ DATA FROM MULTIPLE COLUMNS FROM TABLE
4.SELECT * FROM TABLE1,TABLE2,TABLE3; ----->FOR READ ALL THE DATA FROM ALL 3 TABLES
5.SELECT TABLE1.COLUMN , TABLE2.COLUMN FROM TABLE1,TABLE2; ----->FOR READ COLUMNS FROM DIFF TABLES,
6.SELECT DISTINCT COLUMN_NAME FROM TABLE_NAME; -----> DISTINCT IS USE FOR READ ONLY UNIQUE VALUES

NOTE:- DQL ME ATA SIRF EK HI COMMAND HAI "SELECT" BUT SELECT COMMAND KE CLAUESE BOHOT ZYADA HAI.

```
CREATE DATABASE db_name;
USE DATABASE db_name;
```

```
CREATE DATABASE prit;
USE DATABASE prit;
```

FOR CREATING DATABASE

DATABASE CREATE KRNA IS VERY EASY JUST WRITE CREATE DATABASE AND THEN DATABASE NAME AND LAST ME SEMICOLON AND WRITE USE AND DATABASE NAME FOR GO TO THAT DATABASE AND WORK IN THAT DATABASE

```
CREATE TABLE tb_name (
    column1 datatype constraint,
    column2 datatype constraint,
    column3 datatype constraint,
    column4 datatype constraint,
    column5 datatype constraint
);
```

```
CREATE TABLE users (
    id INT PRIMARY KEY,
    email VARCHAR(100) UNIQUE,
    name VARCHAR(100) NOT NULL,
    age INT CHECK (age >= 18),
    city VARCHAR(50) DEFAULT 'India'
);
```

FOR CREATING TABLE

TABLE CREATE DATABASE KE JITNA EASY NAHI ISME TABLE BANATE VKT NAME COLUMN BHI DEFINE KRNI PADTI HAI OR COLUMN KE NAME , DATATYPE OR BEHAVIOUR (CONSTRAINT) BHI SATH ME DEFINE KRNA PADTE HAI KISI BHI COLUMN KA DATATYPE EK HI HOTA HAI PR CONSTRINTS EKSE ZYADA DEFINE KR SKTE HAI HUM

DATATYPES

- NUMERIC : INT, BIGINT, SMALLINT, DECIMAL(p,s),FLOAT
  - INT ---->NORMAL INTEGER
  - BIGINT ---->BIG INTEGER
  - SMALLINT ---->SMALL INTEGER
  - DECIMAL(p,s)---->EXACT DECIMAL
  - FLOAT ---->APPROX DECIMAL
- STRING : CHAR(n), CHAR(n), TEXT
  - CHAAR(N) ----> ISME HUMNE JITNE VALUE PUT KRI UTNI HI STORE HOGI
  - VARCHAR(N)---->ISME HUM JITNA USE KRENGE UTNA STORE HOGA
- DATE&TIME : DATE, TIME, DATETIME, TIMESTAMP
- BOOLEAN : BOOLEAN (TRUE & FALSE ME ANSWER)
- BINARY : BLOB(IMG,FILES) , VARBINARY (BINARY DATA STORE)
- ENUM : ENUM (FIXED OPTION PROVIDE KRE)
- JSON : DATA JSON(JSON DATA STORE KRVA MODERN DB)

CONSTRAINTS

- PRIMARY KEY — PRIMARY KEY VALI COLUMN UNIQUE OR NOT NULL HOTI HAI
- FOREIGN KEY — DO TABLE KO CONNECT KRTA HAI PERENT CHILD JESE
- UNIQUE — DUPICATE VALUE ALLOW NHI KRTA JESE EMAIL,NUMBER
- NOT NULL — EMPTY VALUE ALLOW NHI KRTA
- CHECK — CONDITION LAGATA HAI JO CONDITION TRUE HO WHI VALUE LEGA
- DEFAULT — DEFAULT VALUE SET KRKE DETA HAI
- AUTO_INCREMENT — AUTOMATIC NUMBER GENERATE KRKE DETA HAI

KEYS

1. PRIMARY KEY :

-- PRIMARY KEY HAS ALWAYS TAKEN THE UNIQUE AND NOT NULL VALUES IN IT BASICALLY YE HAR ROW KO UNIQUELY IDENTIFY KRNE KE LIYE HAI

-- PRIMARY KEY SE HUM KI BHI COLUMN KO DUSRE COLUMN ME FORGIN KEY KE SATH REFREANCE KR SKTE HAI

--METHOD OF WRITING:

```
user_id INT PRIMARY KEY
```

<---- USER ID KE SATH ISLIYE HAI KYUKI USER_ID HAMESA DIFF HI HOTI HAI OR NULL NAHI HO SKTI

2. FOREIGN KEY :

-- FOREIGN KEY HAMESHA PRIMARY KEY SE REFRENCE LETA HAI OR DO TABLES KO USI KEY SE CONNECT KRTA HAI

-- EX. AGAR EK TABLES HAI USERS TO USME USER_ID "PK" HAI OR DUSRA TABLE HAI ORDERS OR HME USME BHI US USER_ID KI JARURT HAI TO FOREIGN KEY USERS TABLE KI PRIMARY KEH KO REFRANCE KREGA IS TABLE KE LIYE OR VAHA SE TABLE CONNECT KREGA

--METHOD OF WRITING

```
FOREIGN KEY id REFRENCE users(user_id)
```

<--- FOREIGN KEY USERS TABLE KI USER_ID SE REFRENCE LE RHA HAI SO ID = USER_ID KEH SKTE HAI HUM

3. UNIQUE KEY:

-- ACTUAL ME YE CONSTRAIN'T HI HAI MTLB YE UNIQUE VALUE HI FILL KRNE DETA HAI SAME VALUE NAHI CHALEGI

--METHOD OF WRITING

```
email VARCHAR(100) UNIQUE
```

<-- YE EMAIL HAMESHA UNIQUE HI CHHAIYE HOGA REGISTER KRNE KE LIYE SO ...

4. CANDIDATE KEY

-- JO KEY PRIMARY KEY BAN SKTI HAI BUT BANI NAHI OR WO UNIQQUE KEY REH GAI USSE HUM CANDIDATE KEY KEHTE HAI

--EXAMPLE

```
email VARCHAR(100) UNIQUE
phone VARCHAR(15)  UNIQUE
```

<-- YE DONO UNIQUE KEY HUM PRIMARY KEY KI JGH BHI RKH SKTE HAI BUT RKHA NHI SO...

5. SUPER KEY

-- UNIQUE COMIBINATION JESE ID BHI UNIQUE AND ID+NAME BHI UNIQUE TO USSE HUM SUPER KEY KAHENGE
--ISKA KOI SYNTAX NAHI HOTA YE CHIZE SIRF OR SIRF CONCEPT ME HAI BAKI REAL LIFE ME ISKA US QUERY ME NHI HOTA SO CHILLLL

6. COMPOSITE KEY

--JISME MULTIPLE COLUMN MILKE PRIMARY KEY BANATE HAI USSE HUM COMPOSITE KEY KEHTE HAI

--METHOD OF WRITING

```
CREATE TABLE order_items (
   order_id INT,
   product_id INT,
   PRIMARY KEY (order_id, product_id)
);
```

<--IS ME ESA HAI KI AGAR HUM KUCH PRODUCT ORDER KR RHE HO TO WO PRODUCT EK ORDER ME REPEAT NHI HOGA
<--INDIRECCT EK ORDER ME SAME PRODUCT REAPT NHI HOGA

7. ALTERNATE KEY

--JO CANDIDATE KEY PRIMARY KEY NAHI BANI USSE HUM ALTERNATE KEY KEHTE HAI

--METHOD OF WRITING

```
CREATE TABLE users (
   id INT PRIMARY KEY,
   email VARCHAR(100) UNIQUE
);
```

<--ISME EMAIL ALTERNATE KEY HAI BCUZ WO BHI HUM AS A PRIIMARY KEY LE SKTE THE BUT ID KO LE LIYA TO AB NHI LE SKTE NA

8. SURROGATE KEY

--JO KEY ARTIFICAL HO I MEAN JO AUTO GENERATED HO

--METHOD OF WRITING

```
id INT AUTO_INCREMENT PRIMARY KEY
```

<--ISME ID KHUD GENERATE HO RHI HAI SO ESI KEY KO SUM SURROGATE KEY KEHTE HAI

9. NATURAL KEY

--JO KEY REAL WORD DATA SE AATI HO USSE NATURAL KEY KEHTE HAI
--EX. ADHAR PAN NUMBER, EMAIL ETC..

SYNTEX

DDL COMMANDS

```
CREATE DATABASE db_name;                    <---create database with or without condition
CREATE DATABASE IF NOT EXIST db_name;

use db_name;                                <---use database

CREATE TABLE table_name (                   <---create table
   column1 datatype,constraints
   column2 datatype,constraints
   FOREIGN KEY column_name REFRENCE table_name(pk_name)
);
```

```
ALTER TABLE table_name                      <--ADD COLUMN
ADD column_name datatype;

ALTER TABLE table_name
ADD (                                       <--ADD MULTIPLE COLUMNS
   col1 INT,
   col2 VARCHAR(50)
);

ALTER TABLE table_name                      <--MODIFY COLUMN (datatype change)
MODIFY column_name VARCHAR(200);

ALTER TABLE table_name                      <--RENAME COLUMN
RENAME COLUMN old_name TO new_name;

ALTER TABLE table_name                      <--DROP COLUMN
DROP COLUMN column_name;

ALTER TABLE table_name                      <--ADD CONSTRAINT
ADD CONSTRAINT pk_name PRIMARY KEY (column_name);

ALTER TABLE orders
ADD CONSTRAINT fk_user                      <--ADD FOREIGN KEY
FOREIGN KEY (user_id) REFERENCES users(user_id);

ALTER TABLE table_name                      <--DROP CONSTRAINT
DROP CONSTRAINT constraint_name;
```

```
DROP DATABASE db_name;                      <--DROP DATABASE

DROP TABLE table_name;                      <--DROP TABLE

DROP TABLE IF EXISTS table_name;            <--DROP TABLE WITH CONDITION
```

🔴 6. TRUNCATE (DELETE DATA FAST ⚡)

SQL

```
TRUNCATE TABLE table_name;
```

👉 Key points:

- Data delete ✔
- Structure safe ✔
- Faster than DELETE ✔
- No rollback (mostly)

➤ RENAME TABLE

SQL

```
RENAME TABLE old_name TO new_name;
```

NOTE:– AGAR RENAME OR DROP TABLES KE LIYE HO DIRECT HUM DROP OR RENAME COMMAND USE KR SKTE HAI BUT AGAD WO KISI COLUMN KE LIYE KRNA HO TO HME "ALTER TABLE" USE KRKE HI USE KRNA HOGA

DML COMMAND

```
INSERT INTO table_name (col1, col2)         <--INSERT ROWS IN TABLE
VALUES (value1, value2);

INSERT INTO table_name                      <--INSERT ROW IN TABLES WITHOUT SPECIFY THE COLUMN
VALUES (value1, value2);                       BUT YOU HAVE TO PUT VALUE IN SAME ORDER OF COLUMN

INSERT INTO users (id, name, age)
VALUES
(1, 'Prit', 22),                            <--INSERT MULTIPLE ROW IN COLUMNS
(2, 'JAY', 21),
(3, 'JAY', 22);

INSERT INTO new_users                       <--INSERT DATA FROM ANOTHER TABLE
SELECT * FROM users;

INSERT INTO new_users (id, name)            <--INSERT SPECIFIC COLUMN FROM ANOTHER TABLE
SELECT id, name FROM users;

INSERT IGNORE INTO users (id, name)
VALUES (1, 'Prit');                         <--DUPLICATE VALUES STORE KRNE DEGA
```

```
UPDATE table_name
SET column_name = value
WHERE condition;

   EX:
      UPDATE users                          <--only for one column jaha id 1
      SET name="prit"                          hoga waha prit name ho jayega
      where id=1 ;
   EX 2:                                    <--multiple columns ke liye
      UPDATE users
      SET name="prit" , age = 22
      where id = 1;

UPDATE users u                              <--update using another table
JOIN orders o ON u.id = o.user_id              isme dusre table se refrence liya
SET u.status = 'active'                        jata hai value change krne ke liye
WHERE o.amount > 1000;

UPDATE users
SET age = age + 1;                          <--increment bhi kr skte hai or dusri calculation bhi
```

NOTE: BINA WHERE CONDDITION KE CHALAYA TO PURA DATA MODIFIED HOJAYGA SO ALWAYS USE WHERE WITH IT

```
DELETE FROM table_name                      <--DELETE PURI ROW HI DELETE KRTA HAI
WHERE condition;

DELETE FROM table_NAME;                     <--WITHOUT CONDITION VALA DELETE PURE TABLE KI VALUES DELETE KREGA

DELETE u
FROM users u
JOIN orders o ON u.id = o.user_id           <--DELETE WITH JOIN
WHERE o.amount < 100;
```

```
INSERT INTO users (id, name)                <--THIS IS UPSERT(MERGE) AGAR DATA HOGA TO UPDATE
VALUES (1, 'Prit')                             KREGA NAHI HOGA TO NAYA DATA INSERT KREG
ON DUPLICATE KEY UPDATE name = 'Updated Prit';
```

NOTE: UPDATE AND DELETE ALWAYS USE WITH WHERE CONDITION WITHOUT THIS IT WILL AFFECT WHOLE COLUMN

--> DELETE SE PURI ROW DELETE HOGI BUT IF TUME SIRF KOI EK PLACE DELETE KRNI HO TO USE UPADTE AND SET VALUE AS NULL

-->DELETE ROW LEVEL OPRATION HAI OR UPDATE COLUMN LEVEL